

SE 317: Module 3 Lab

Part 1: Bubble Sort Testing

Write a java program that implements the bubble sort algorithm. Your program will take an array of unsorted integers (at least 7 integers) and sort them, then print the old and the new arrays in different lines. Do the following steps:

- a) Write a faulty program (include any fault of your choice in your code). Make sure your code still compiles successfully.
 - i. Write down your source code in Java. The code output will display the original (input) array and the new array on different lines. **Make sure all your code is clearly commented.**
 - ii. Compile your code and take a screenshot.
- b) Write **two** tests that do NOT reveal the fault.
 - i. Each test case will include an input array of your choice and the expected output array.
 - ii. Run your code using the test array as input and take a **screenshot** of the input and actual output of each test case.
- c) Write two tests that do reveal the fault.
 - i. Each test case will include an input array of your choice and the expected output array
 - ii. Run your code using the test array as input and take a **screenshot** of the input and actual output of each case.
- d) Identify and remove the fault.
 - i. Explain your correction in the comments of your source code.
 - ii. Run the test cases of **part c** again after fixing the fault.
 - iii. Take a **screenshot** of the input and actual output of each case.

Part 1 Submission:

- 1) Submit your source code (.java files with appropriate names) in ne folder named “bubble sort”
- 2) Submit a Word (or PDF) document showing **all your source code** with clear comments
- 3) Include all screenshots explained above in parts a, b, c, d (=7 screenshots).
- 4) In your own language, explain what the fault was in your algorithm, and how you fixed it
- 5) Explain why the first two tests (in part b) failed to reveal the error while the tests in part c identified the errors successfully.

Part 2: Quick Sort Testing

Repeat the same steps and submissions as in part 1 using Quick Sort

Part 3: Merge Sort Testing

Repeat the same steps and submissions as in part 1 using Merge Sort

Put all your files in one zip file and upload it to the Canvas.